#HW 7

Graded

Student

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Total Points

100 / 100 pts

Question 1

(no title) **10** / 10 pts

✔ **– 0 pts** Correct

– 2 pts del T is same in both rankine and fahrenhit scales.

– 2 pts 6 inches is 0.5 ft

– 5 pts Q = K A del T / L.

– 3 pts check the calculation properly.

– 1 pt total heat loss in 10 hours not calculated

– 10 pts not answered

Question 2

(no title) **15** / 15 pts

✔ **– 0 pts** Correct

– 2 pts Q total = Q convection + Q radiation ; step missed

– 2 pts check the calculation properly

– 15 pts question hasn't been answered

Question 3

(no title) **15** / 15 pts

✔ **– 0 pts** Correct

– 3 pts W = m (g/gc)

– 3 pts m = PV/RT

– 2 pts check the units/ calculation properly

– 10 pts didnt answer the question properly

Question 4

(no title) **15** / 15 pts

- ✔ **– 0 pts** Correct
- **– 2 pts** calculation error
- **– 5 pts** Q = m Cp del T
- **– 5 pts** incomplete calculation

Question 5

(no title) **15** / 15 pts

- ✔ **– 0 pts** Correct ; 634 K or 361 C
- **– 3 pts** calculation error

Question 6

(no title) **15** / 15 pts

- ✔ **– 0 pts** Correct
- **– 5 pts** incomplete solution
- **– 3 pts** Calculation mistake/ error.

Question 7

(no title) **15** / 15 pts

- ✔ **– 0 pts** Correct
- **– 3 pts** calculation error in work done
- **– 2 pts** calculation error in Q
- **– 2 pts** part b calculation error/ not completed.

CHE 201HW 7

Due: 11.59pm on 10/20/2025

1. (10 points)
Problem 2.13.35

2. (15 points)
Problem 2.13.41

3. (15 points)
Problem 3.21.71

4. (15 points)
Problem 3.21.79

5. (15 points)
Problem 3.21.84

6. (15 points)
Problem 3.21.90

7. (15 points)
Problem 3.21.91

1. $\dot{Q} = \dfrac{-kA(T_2 - T_1)}{x}$

$x = 6in = 0.5ft$

$A = 20 \times 10 ft^2$
$= 200ft^2$

$k = 0.04\ Btu/h \cdot ft\ ^{\circ}R$

$\dot{Q} = \dfrac{-0.04 \cdot (200) \cdot (40-68)}{0.5}$

$\dot{Q} = 448\ Btu/h$

$E = \dot{Q} \times t =$
$448 \times 10 = \boxed{4480\ Btu}$

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2. $T_{grill} = 47^{\circ}C + 273 = 320K$
$T_{surr} = 27^{\circ}C + 273 = 300K$
$\varepsilon = 0.93$
emissivity of grill

stefan boltman
constant
$\sigma = 5.67 \times 10^{-8} \dfrac{W}{m^2 K^4}$

$h = 10 \dfrac{W}{m^2 K}$
heat transfer cofficient

2A. $q_c = h(T_g - T_s)$

Heat loss per unit Area by convection

$q_c = 10(320 - 300)$

$q_c = 200 \frac{W}{m^2} = \frac{200}{1000} \frac{kW}{m^2}$

$$\boxed{q_c = 0.2 \frac{kW}{m^2}}$$

Heat loss per unit Area by radiation

$q_R = \varepsilon \sigma (T_g^4 - T_s^4)$

$q_R = 0.93 (5.67 \times 10^{-8})(320^4 - 300^4)$

$q_R = 125.8 \, W/m^2 = 125.8/1000 = 0.1258 \frac{kW}{m^2}$

Net rate of heat transfer

$q_T = q_C + q_R$

$q_T = 0.2 + 0.1258$

$$\boxed{q_T = 0.3258 \, kW/m^2}$$

3. $V = (110 \cdot 60 \cdot 25)\,ft^3 = 165000\,ft^3$

$T = 70^\circ F = 70^\circ F + 459.67 = 529.67^\circ R$

$P = 1\,atm = 14.696\,PSI = 2116.224\,lbf/ft^2$

$R = 1545\,ft \cdot lbf/lb\,mol \cdot {}^\circ R$ $\qquad g = 32.0\,ft/s^2$

$M = 28.97$

$R/M = 1545/28.97 = 53.33$

$$M = \frac{PV}{RT} = \frac{2116.224 * 165000\,ft^3}{53.33 \cdot 529.67^\circ R} = 12361.42$$

$W_{air} = m \cdot g \qquad 1\,lbf = 32.174\,lb \cdot ft/s^2$

$12361.42\,lb \cdot 32\,ft/s^2 = 395565.67\,lb \cdot ft/s^2$

$$= \frac{395565.67\,lbf \cdot ft/s^2}{32.174\,lbf \cdot ft/s^2} = 12294.57$$

$\boxed{Weight_{air} = 12294.57}$

$\boxed{Mass_{air} = 12361.42\,lb}$

4. $M = 2kg \quad V_1 = 2m^3 \quad P_1 = 1atm$

$k = 1.35 \quad V_2 = 2V_1 \quad R = 8.314$

$$P_1 V_1 = mRT \rightarrow T_1 = \frac{P_1 V_1}{m \cdot R}$$

$$T_1 = \frac{\left(1\,bar \cdot \frac{10^5\,Pa}{1\,bar} \cdot \frac{1\,kPa}{10^3}\right) \cdot 2m^3}{2 \cdot 8.314/32} = 384.893\,K$$

$$\frac{V_2}{V_1} = \frac{T_2}{T_1} \rightarrow T_2 = \left(\frac{V_2}{V_1}\right) T_1 = \left(\frac{2(2)\,m^3}{2m^3}\right) 384.893\,K$$

$T_2 = 769.786\,K$

$$c_v = R/k-1 = \frac{0.259}{1.35-1} = 0.742\,kJ/kg \cdot K$$

$$Q = \Delta U + W$$

$$= mc_v(T_2 - T_1) + P_1(2V_1 - V_1)$$

$$= 2kg \cdot 0.742\,kJ/kg \cdot K \,(769.786\,K - 384.893\,K) + 100\,kPa\,(4m^3 - 2m^3)$$

$$= 571.181\,kJ + 200\,kJ$$

$$= \boxed{771.181\,kJ}$$

5. $T_1 = 430^\circ C$ $\quad k = 1.35$ $\quad \dot{Q} = 425\ W$

$m = 5\ kg$ $\quad$ For 10 mins

$Q = \Delta U + \cancel{W}^{0}$ since Rigid tank (volume does not change) $\quad \Delta U = \dot{Q}$

$$C_v = \frac{R}{k-1} = \frac{0.259\ kJ/kg \cdot K}{1.35 - 1} = 0.7423\ kJ/kg \cdot K$$

$$\Delta U = m c_v (T_2 - T_1) = Q \Rightarrow T_2 - T_1 = \frac{Q}{m c_v} \rightarrow$$

$$T_2 = \frac{Q}{m c_v} + T_1 = \frac{-255}{5 \cdot 0.7423} + 703 = 634.4\ K$$

$$T_2 = 634.29 - 273 = \boxed{361.29^\circ C}$$

$\dot{Q} = \frac{Q}{t} = \dot{Q} \cdot t = -425 \cdot 600 = -255{,}000\ J = -255\ kJ$

Negative

Heat Removed

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6. $T_1 = 560^\circ R$ $\quad P_1 = 18\ lbf/in^2$

$V_1 = 0.29\ ft^3$

$T_1 = 560^\circ R = 100^\circ F$

$C_p = 0.240\ Btu/lbm$ $\quad$ Find at 16.20 Zybook

$C_v = 0.172\ Btu/lbm$

$R = C_p - C_v$

$= 0.240 - 0.172$

$R = 0.068 \text{ Btu/lbm}$

$P_1 V_1 = m R T_1 \rightarrow \quad m = \frac{PV}{RT}$

$$m = \frac{(18 \text{ lbf/in}^2) \cdot \left(0.29 \text{ ft}^2 \cdot \left(\frac{12 \text{ in}}{1 \text{ ft}}\right)^3\right)}{0.068 \text{ Btu/lbm} \cdot \frac{9338.03 \text{ lbf in}}{1 \text{ Btu}} \cdot 560^\circ R}$$

$= 0.02536 \text{ lbm}$

$Q = \Delta U + W$ there no heat transfer so $Q = 0$

$\Delta U = m c_v (T_2 - T_1)$

$W = -1.7$

$Q = m C_p (T_2 - T_1) + W$

$1.7 = 0.02536 \times 0.240 \times (T_2 - 560)$

$$\boxed{T_2 = 839.31^\circ R}$$

7. $P_1 = 0.7\text{ bar}$ $T_1 = 280K$ $P_2 = 11\text{ bar}$

a) $V_1 = 0.262 m^3$ $n = 1.25$

$$P_1 V_1^n = P_2 V_2^n$$

$$V_2 = \left(\frac{P_1}{P_2}\right)^{1/n} V_1 \rightarrow V_2 = \left(\frac{0.7\text{ bar}}{11\text{ bar}}\right)^{1/1.25} (0.262 m^3)$$

$$= 0.02892 m^3$$

$$W_{1-2} = \int_1^2 P\,dV = \frac{P_2 V_2 - P_1 V_1}{1-n}$$

$$= \frac{\left(11\text{ bur} \cdot \frac{10^5 Pa}{1\text{ bar}} \cdot \frac{1 kPa}{1 Pa}\right)(0.02892 m^3) - \left(0.7\text{ bar} \cdot \frac{10^5 Pa}{1\text{ bar}} \cdot \frac{1 kPa}{10^3 Pa}\right)(0.262 m^3)}{1-1.25}$$

$$= -53.91\text{ kJ}$$

$$P_1 V_1 = mRT_1 \rightarrow m = \frac{P_1 V_1}{RT_1}$$

$$m = \frac{(70 kPa \cdot 0.262 m^3)}{\frac{8.314}{44.01}\frac{kJ}{kg \cdot K} \cdot 280K} = 0.3467 kg$$

Now find T_2

$$P_2 V_2 = mRT \rightarrow T_2 = \frac{P_2 V_2}{mR}$$

$$T_2 = \frac{(1100\ kPa)\cdot(0.02892\ m^3)}{0.3467 kg \cdot 8.314/44.01\ \frac{kJ}{kg\cdot K}} = 485.7K$$

look at 15.20 txbook

at 300K $c_v = 0.657\ kJ/kg\cdot K$

$$\Delta U_{1\to2} = mc_v(T_2 - T_1)$$
$$= (0.3467 kg)(0.657 kJ/kg\cdot K)(485.7K - 280K)$$
$$= 46.85\ kJ/kg$$

$$Q_{1\to2} = \Delta U_{1\to2} + \Delta W_{1\to2}$$

$$Q_{1\to2} = 46.85\ kJ/kg + -53.91\ kJ$$
$$= \boxed{-7.06\ kJ}$$

b) at $T_1 = 280K$

$U_1 = 6369\ kJ/kmol$

at $T_2 = 485.7K$

$$U_2 = \frac{T_k - T_1}{T_2 - T_1} = \frac{U_{unknown} - U_1}{U_2 - U_1}$$

$$= \frac{485.7K - 480}{490 - 480} = \frac{U - 12800}{13158 - 12800}$$

$U_2 = 13004\ kJ/kmol$

$$Q_{1\to2} = \Delta U_{1\to2} + W_{1\to2}$$

$$Q_{1\to2} = \frac{m}{M}(U_2 - U_1) + W_{1\to2}$$

$$Q_{1\to2} = \frac{(0.3467 kg)}{(44.01\ kg/kmol)}(13004 - 6369) + (-53.89\ kJ)$$

$$\boxed{= -1.62\ kJ}$$